Project Design Phase
Solution Architecture

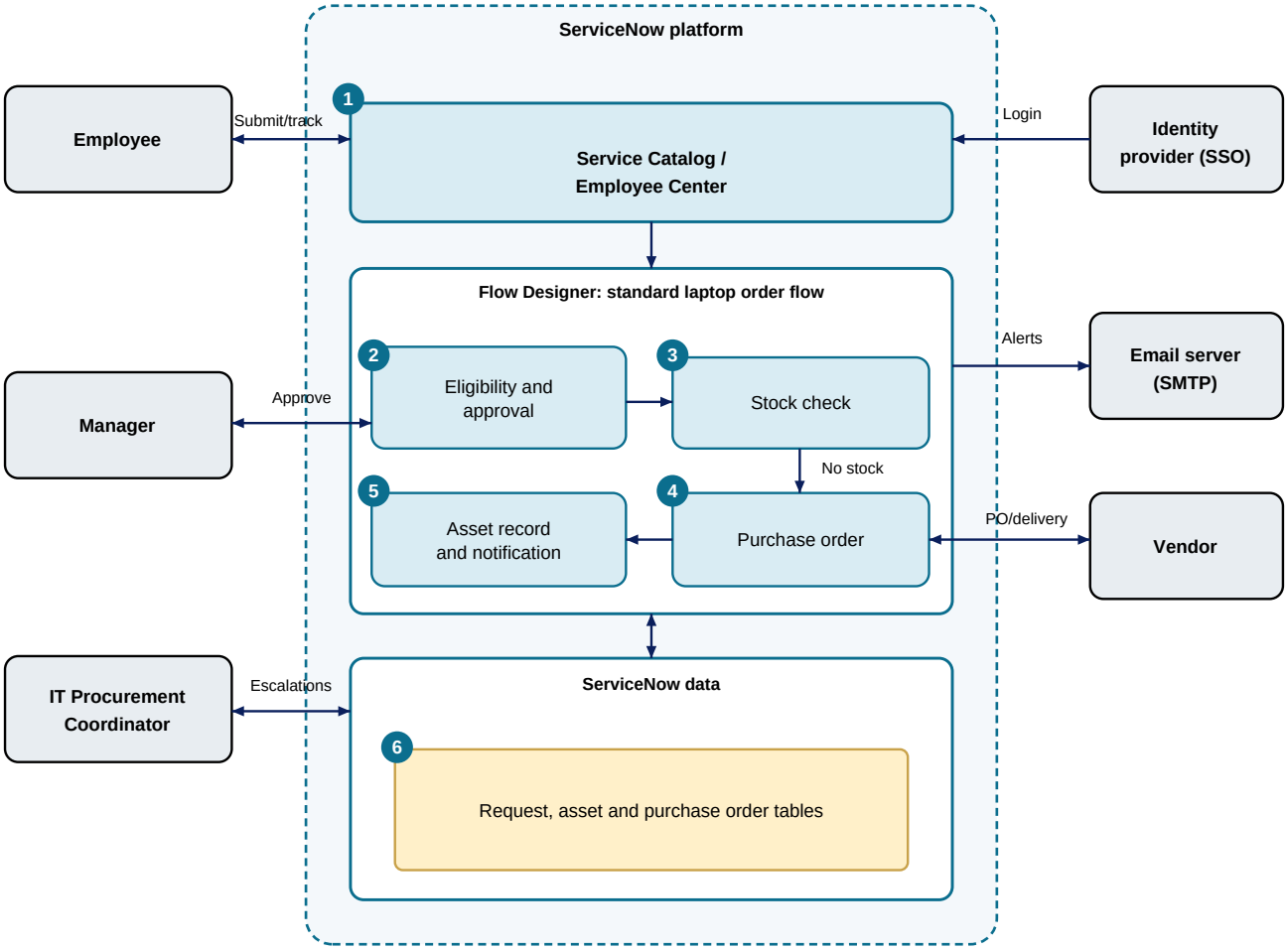
Date	2 October 2026
Team ID	
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a multi-step process that connects business problems with technology solutions. In this project, it illustrates how a standard laptop order moves through ServiceNow, from the employee's request through to delivery.

Goals:

- Identify the most suitable technology to fix the current problem of slow, manual laptop ordering.
- Explain the structure, characteristics and behavior of the solution to the project stakeholders.
- Specify the features, development phases and requirements of the solution.
- Set out the specifications that guide how the solution is defined, managed and delivered.



Architecture Components and Flow:

Each number in the diagram corresponds to a component listed below.

No.	Component	Role in the solution
1	**Service Catalog / Employee Center**	The employee raises a single standard laptop request. Information is auto-filled from their user profile, and the request status can be tracked on the same page.
2	**Eligibility and approval**	Flow Designer validates eligibility. Pre-approved roles are cleared automatically; otherwise a one-click approval task goes to the manager and is escalated if left unanswered.
3	**Stock check**	The flow looks up the asset inventory and holds a laptop for the request whenever one is in stock.
4	**Purchase order**	If inventory runs short, the flow generates a purchase order and routes it to the vendor chosen by region or price.
5	**Asset record and notification**	Once the laptop is delivered, the flow creates the asset record, assigns it to the employee and sends updates throughout the process.
6	**ServiceNow data**	The request, asset and purchase order tables hold the data used by every step, as well as by reports and dashboards.
-	**External systems**	The identity provider manages sign-in, the email server delivers notifications, and the vendor receives the purchase orders.

> Flow Designer coordinates each step within the ServiceNow platform, allowing the employee, manager and coordinator to work from a single system rather than moving the request around through emails and spreadsheets.

Implementation Screenshots (ServiceNow)

Captured from the project demo video: Flow Designer build and a test laptop request.

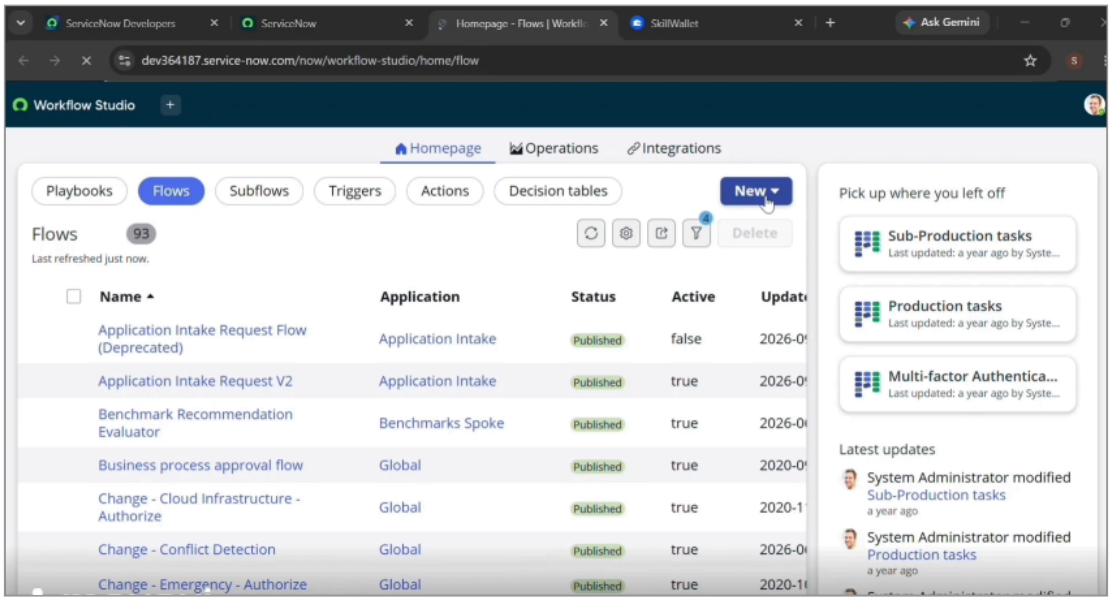


Figure 1: Workflow Studio home page in ServiceNow

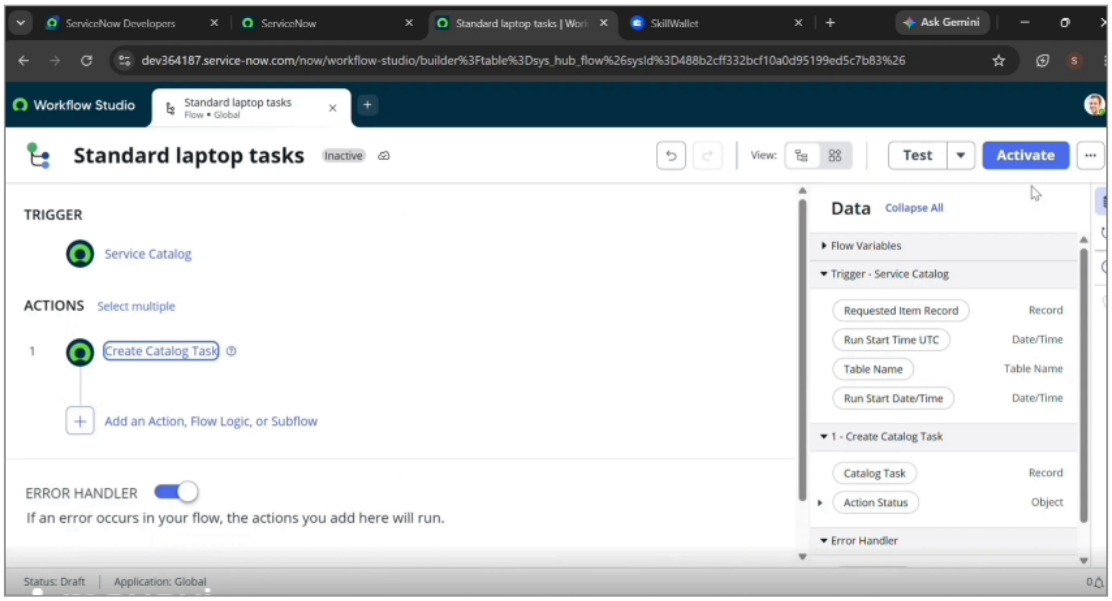


Figure 2: Flow Designer: Service Catalog trigger and Create Catalog Task action

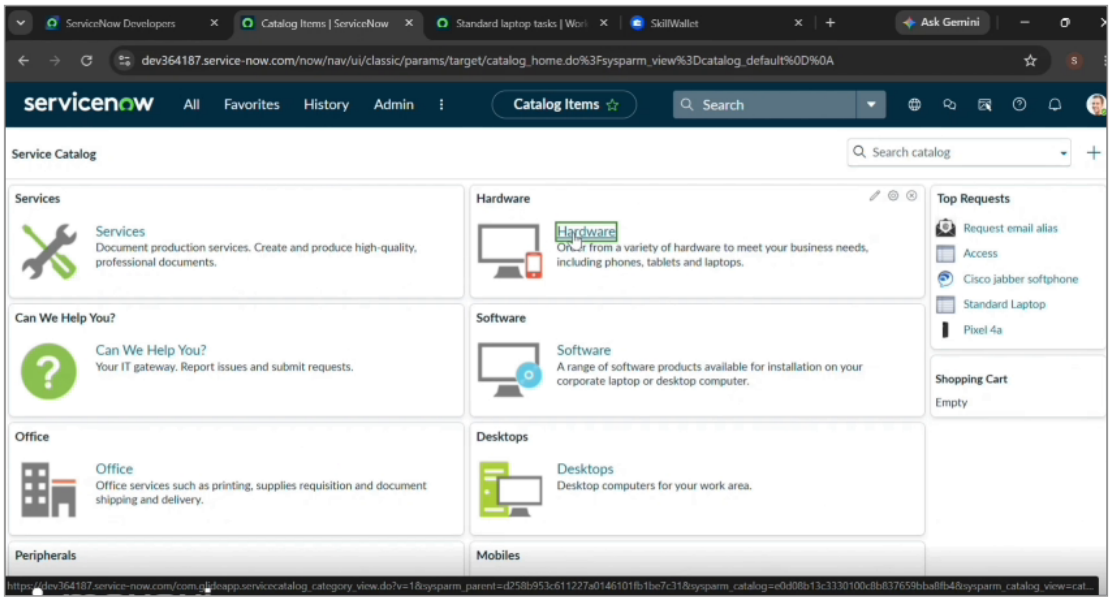


Figure 3: Service Catalog: Hardware category